
Where Should Activities Be Located?

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In 2006 the ports of Hong Kong and Shenzhen shipped one container every second.

YOUR IB VOICE

EXPLAIN THE LOCATION

Fallows claims that neither a stronger Chinese currency nor an end to subsidies would move Shenzhen's work back to the United States. What attracts production to the Pearl River Delta?

10 minutes · Individual · Notes permitted

Keep your response. You will return to it in the closing discussion.

What attracts production to the Pearl River Delta?

Wages in Shenzhen are a fraction of U.S. wages.

Suppliers and parts sit within a mile of the factory.

The government built the zone and waived the duties.

People like Liam Casey know where the factories are.

Workers can be retasked faster than machines.

Why is the work in Shenzhen, and why does it stay?

Back to the future

In 1980 Beijing made Shenzhen a special economic zone beside Hong Kong (harbor + airport).

Nearly every rule that restricted business was changed or removed.

Imported parts entered the processing zones duty-free, and the goods built from them left untaxed



Croquant, *Pearl River Delta Area*, 2007 · Wikimedia Commons, CC BY 3.0.



Workers came young from the countryside
A shift ran twelve hours, six or seven days a week.

Taiwan's model filled the delta with small firms

No. 206

was Foxconn's rank on the Fortune Global 500, as China's largest exporter

Taiwanese entrepreneurs built thousands of small family firms.

Japan and Korea had built national champions instead.

Foreign buyers “don't know where to start.”

| A crowd of small firms needs a guide before a buyer can use it.

Liam Casey sells knowledge of the factories

Casey, from Cork, set up PCH near Shenzhen a year after he first saw Asia.

PCH had 800 employees and sales of about \$125 million in 2006.

His map holds every supplier's GPS position, photographs, and travel times.



Robert Scoble, [Bigger than life](#), 2008 · Wikimedia Commons, CC BY 2.0.

“Supply chain is intellectual property.”

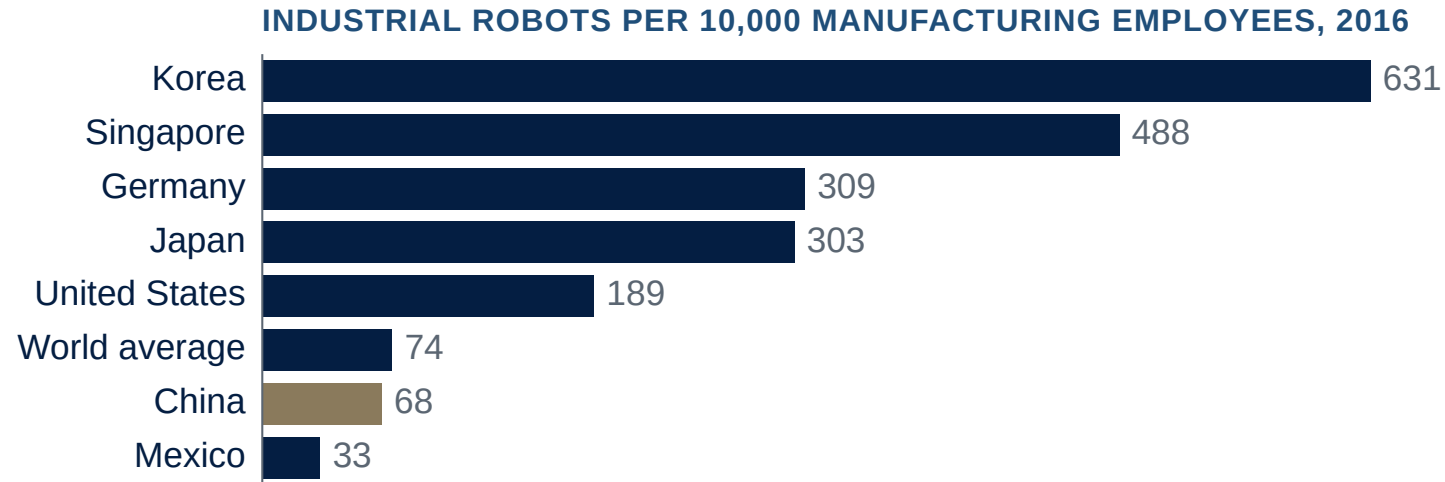
Nine suppliers sit within a mile of the factory



The SEG market stacks seven floors of parts a short drive from hundreds of factories.

Bobbie Johnson, *Inside SEG Plaza*, 2010 · Wikimedia Commons, CC BY-SA 2.0.

A pre-automation era



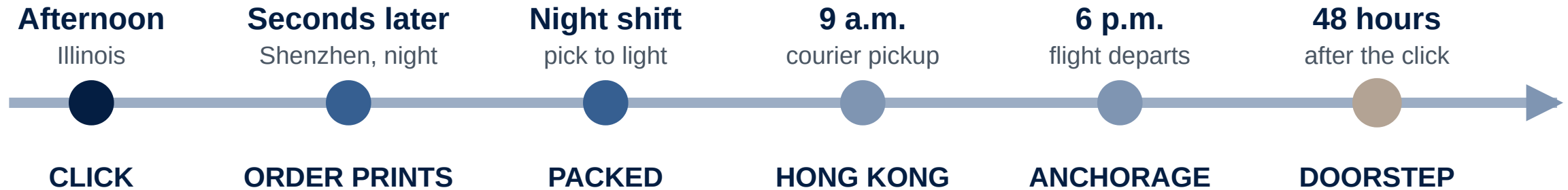
International Federation of Robotics, [World Robotics 2017](#), robot density in 2016.

“Machines need to be reprogrammed. You can have people doing something entirely different next week.”

Shenzhen’s plants stayed short of robots because retasking people was cheaper than reprogramming machines.

■ Cheap labor buys flexibility, and buyers pay for flexibility.

An Illinois order ships from Shenzhen in 48 hours



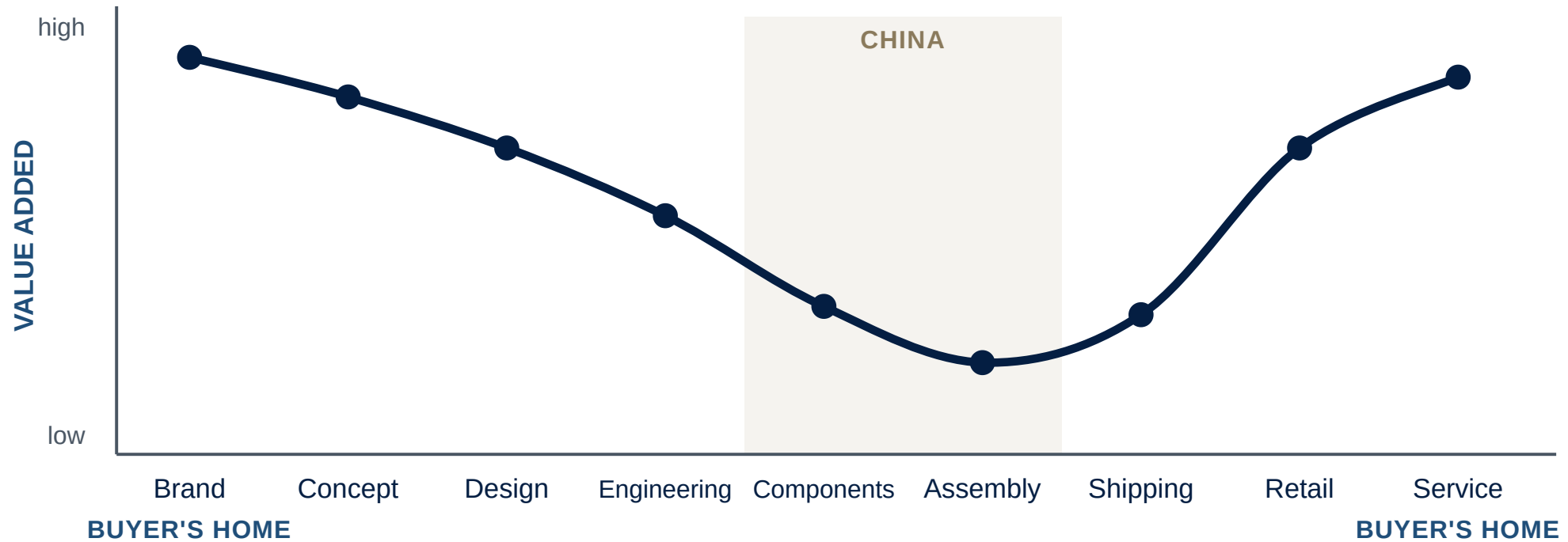
| The night shift packs before the 9 a.m. courier, so the time difference became an asset.

A man in a black t-shirt is riding a blue electric scooter on a city street. He is pulling a black metal cart loaded with several large cardboard boxes. The cart has a blue logo that reads "德邦快递" (Debon Express). In the background, there is a white and blue taxi with the name "康达尔" (Kangdalar) and the number "95128" on its side. The taxi's license plate is "粤B·DG9035". Other people and buildings are visible in the background, creating a busy urban scene.

“People think China is cheap, but really, it’s fast.”

The cluster is fast. Who gets paid for it?

Value sits at the two ends of the 'smiley' curve



China's work sits in the middle stages, and brand, design, retail, and service stay with the buyer.

A \$1,000 laptop leaves \$30 to \$40 in China

4%

or less of a \$1,000 laptop's price stayed with the Chinese factory and its workers

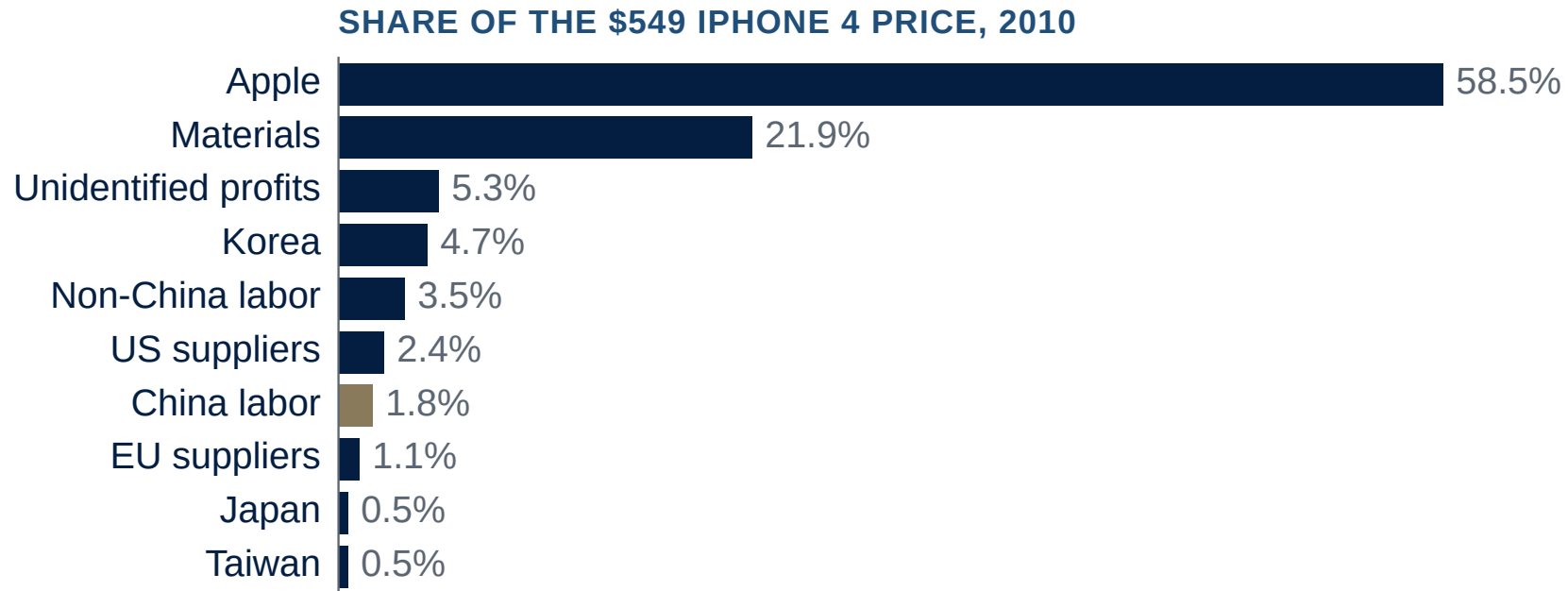
Intel and Microsoft collected about \$300.

The makers of the screen, the disk, and the other components got about \$150 apiece.

The retailer kept under \$50.

Chinese workers earning \$1,000 a year help U.S. designers and retailers earning \$1,000 a week earn more.

Apple kept 58 percent of the iPhone's price



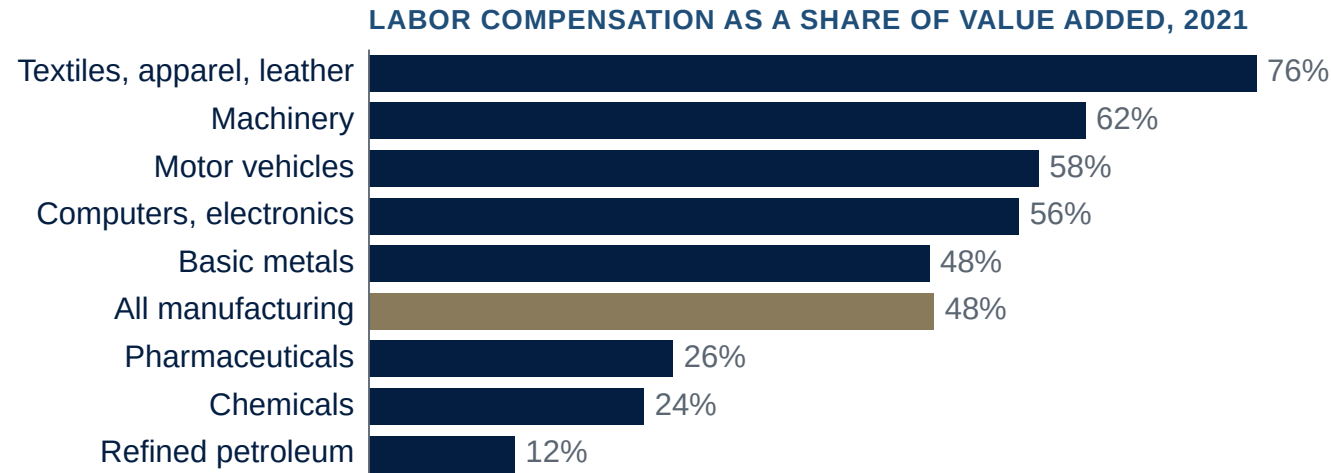
Kraemer, Linden, and Dedrick, *Capturing Value in Global Networks: Apple's iPad and iPhone*, UC Irvine, 2011, Table 1.

Researchers at UC Irvine traced every dollar of the 2010 iPhone 4's \$549 price.

Assembly earned little because most could do it, and design and brand earned most because nobody else could.

Why do different stages of a product go to different countries?

Each industry leans on one input above the others



OECD, [STAN Database for Structural Analysis](#), 2025 edition: United States, 2021, compensation of employees as a share of value added.

Casey turned away a Pittsburgh steel firm: “The product’s too heavy, and you’ve probably already automated the process.”

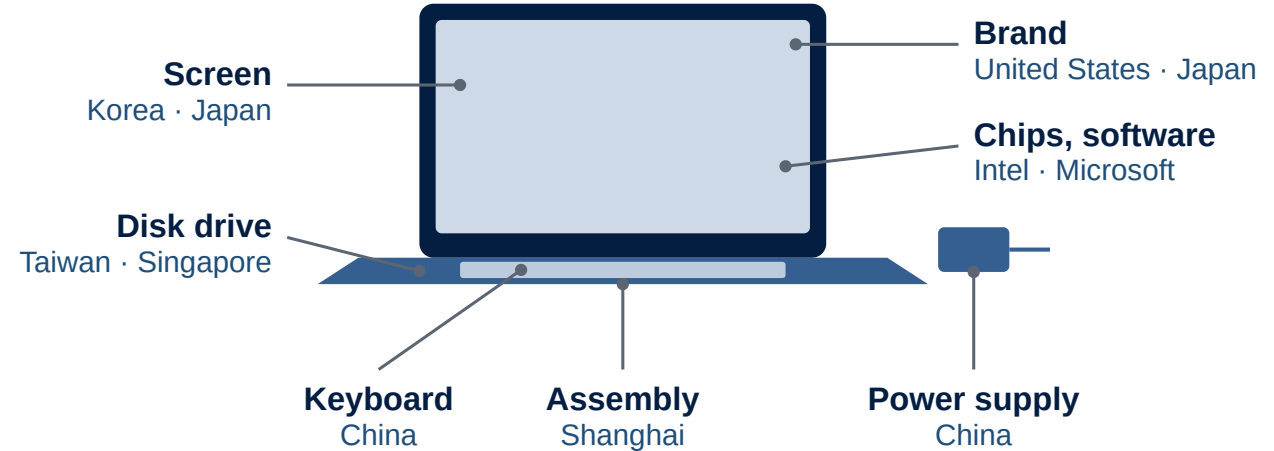
Lower skilled worker Wages matter most to the industries at the top of the list and hardly at all to those at the bottom.

Factor intensity is the input an activity uses most heavily, whether labor, machines, engineers, energy, or land.

Each part of the laptop came from a different place

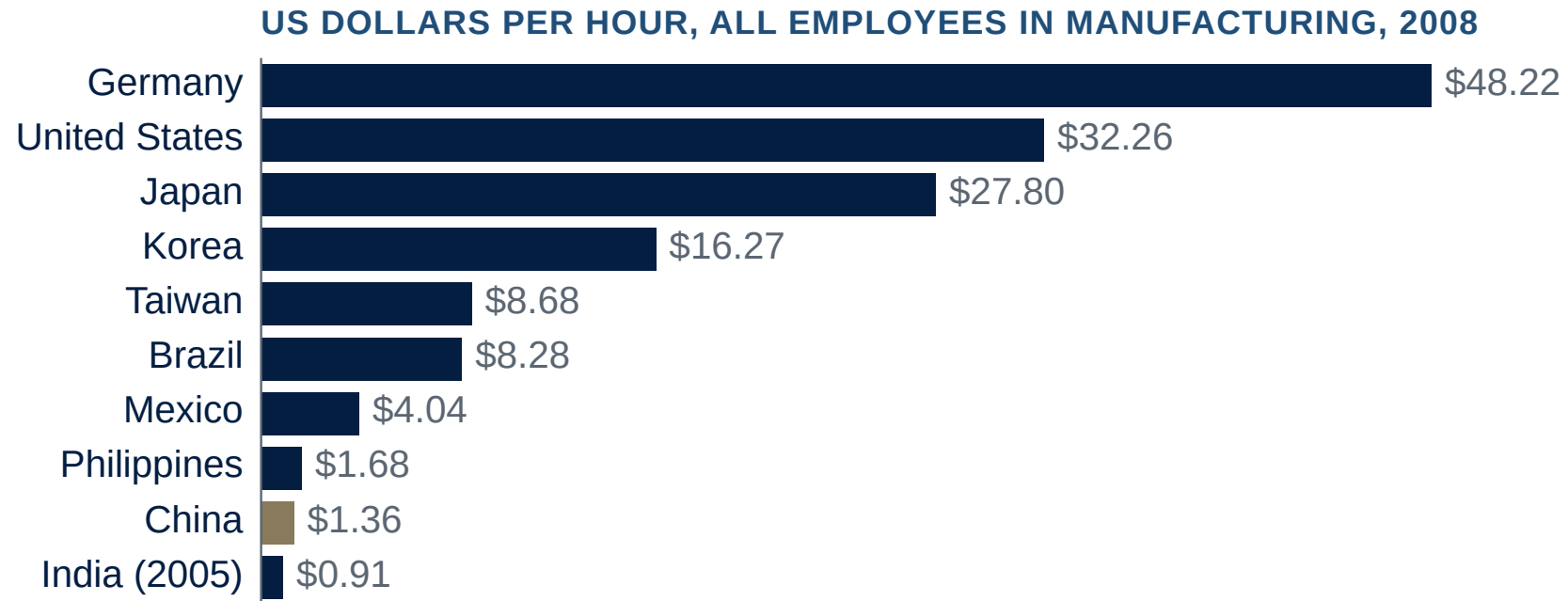
Screens and drives need costly plants, and came from Korea, Japan, Taiwan, and Singapore.

Keyboards, power supplies, and assembly need workers, and were done in China.



Parts of one product lean on different inputs, so they need not be made in one place.

An activity goes where its heaviest input is cheap

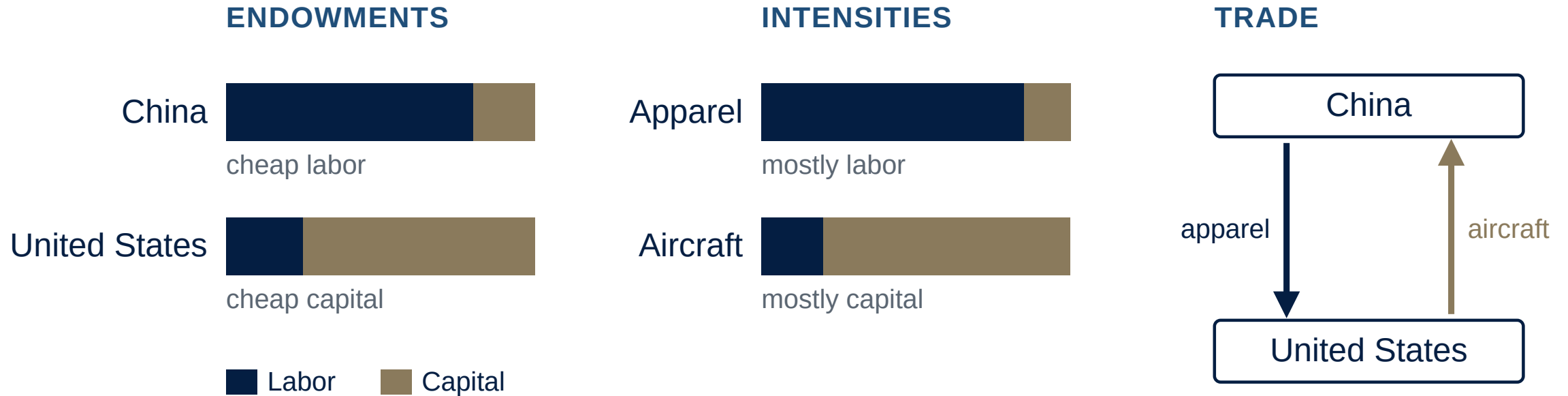


U.S. Bureau of Labor Statistics, [International Comparisons of Hourly Compensation Costs in Manufacturing, 2008](#), 2010; China and India are BLS estimates.

A stage that is mostly labor cost a twentieth as much in China, and a stage that is mostly machines cost about the same anywhere.

A place has a comparative advantage in the activities that use its abundant input most heavily, because that input is cheaper there.

Trade follows endowments and intensities



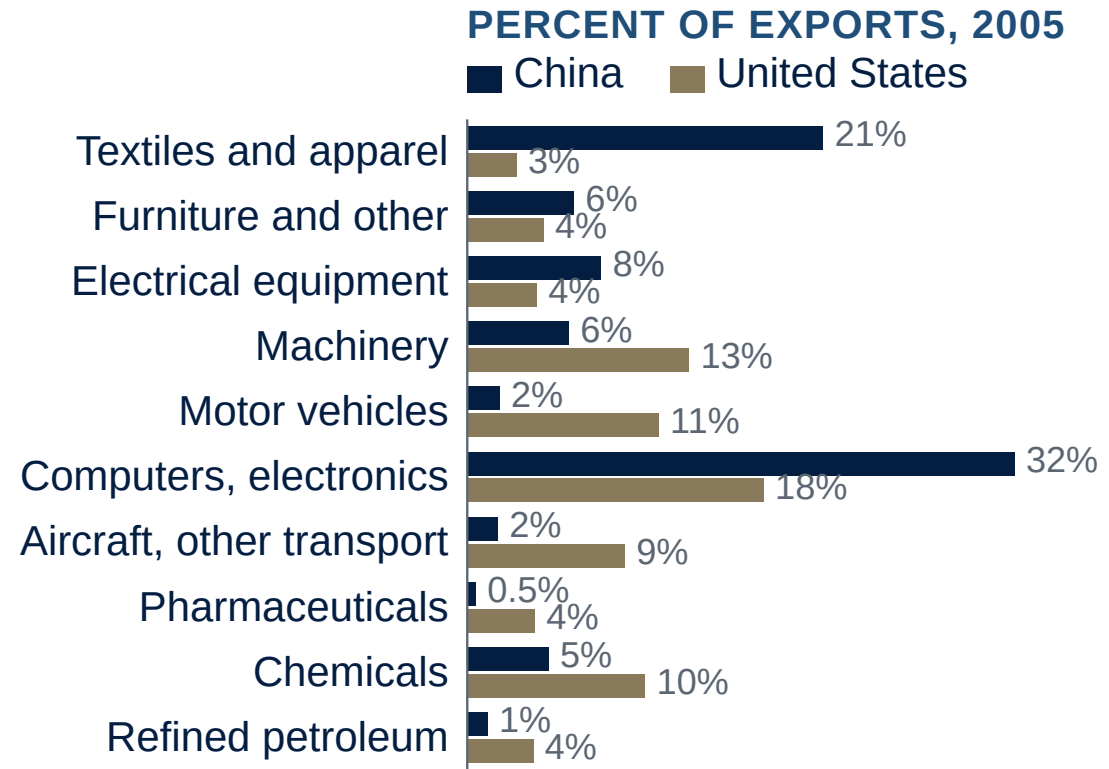
China has many workers per machine and the United States many machines per worker, so apparel is cheaper to make in China and aircraft in the United States.

Heckscher and Ohlin predicted that a country exports the goods that use its abundant input most heavily and imports those that use its scarce one.

China exported the goods that use the most labor

China's exports crowded into the industries where labor matters most, and U.S. exports into those where machines and engineers do.

Electronics sits on both sides because China did the assembly stage and the United States the chips.



OECD, *Trade in Value Added*, 2025 edition: gross exports by industry, 2005; industries in the order of the labor-share chart.

The theorem holds for stages of a product as much as for industries, which is why the question is where an activity goes.

Assembly went where labor was cheapest

90%

of brand-name laptops were built in mainland China by Quanta, Compal, Inventec, Wistron, or Asustek

Inventec's Shanghai plant turned out 30,000 notebooks a day.

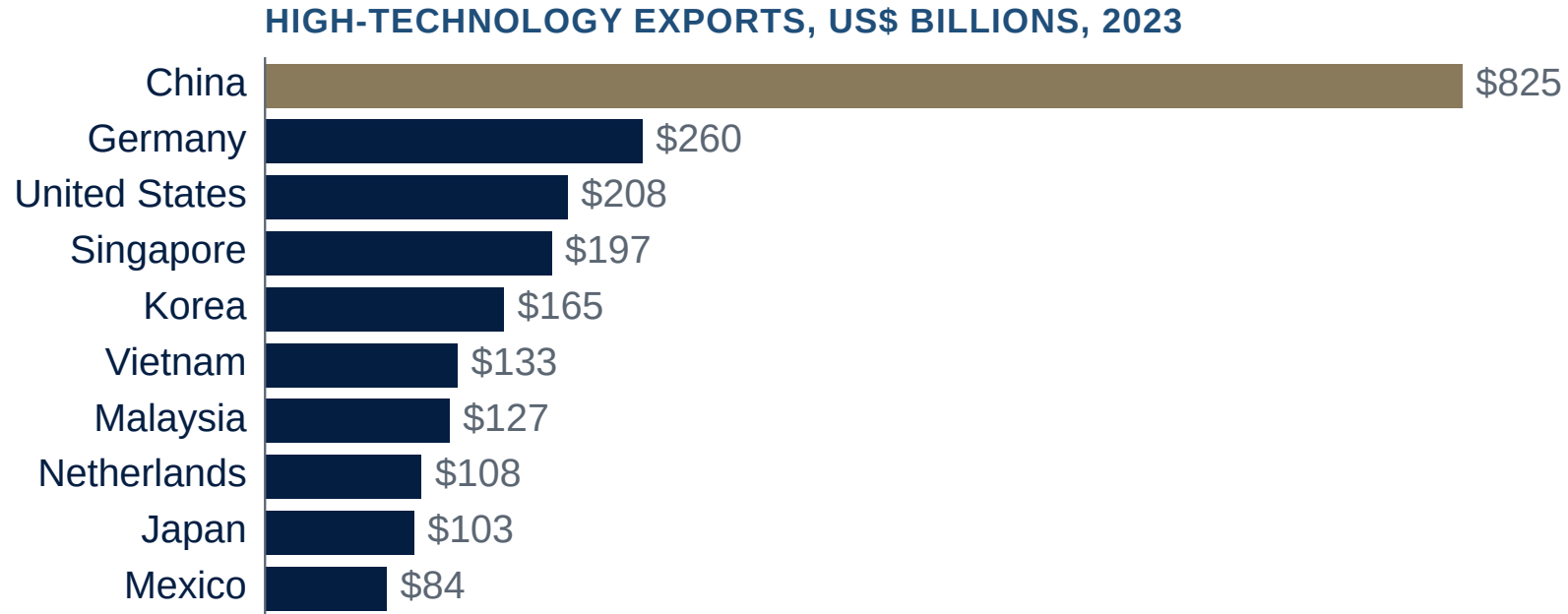
Three rival brands came off one line.

Bare boards and chips came in, and notebooks went out.

■ The labor-intensive stage went to the labor-abundant place.

What holds the cluster, and what moves it?

Density itself becomes an input



World Bank, [World Development Indicators](#), high-technology exports (UN Comtrade), 2023.

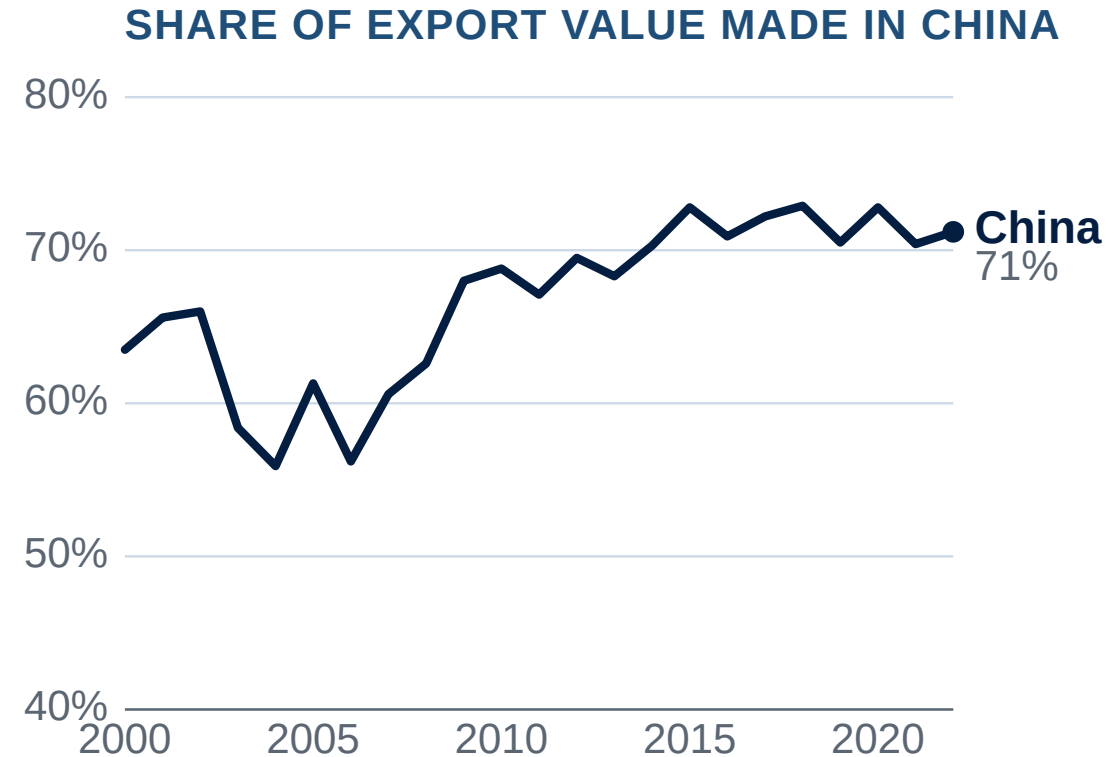
No policy can copy a cluster this size, because the advantage is the other firms already there.

Agglomeration economies are cost and speed advantages a firm gets only because other firms and workers are already nearby.

The parts followed the assembly lines to China

In the processing-trade years China's plants assembled imported parts.

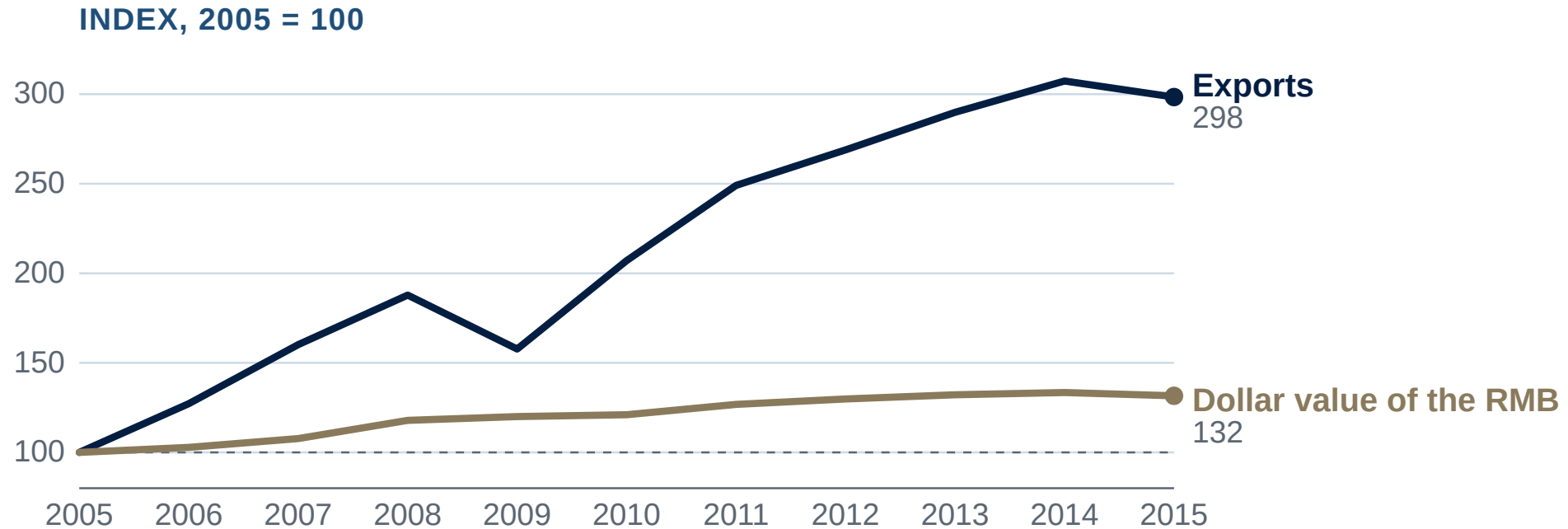
A decade later the parts were made next door, and a supplier base is slower to rebuild than an assembly line.



OECD, [Trade in Value Added](#), 2025 edition: domestic value added in gross exports, computer, electronic and optical products, China.

As suppliers moved next to the assembly lines, more of each export dollar stayed.

The RMB rose a third and exports tripled



World Bank, [World Development Indicators](#): merchandise exports (WTO) and official exchange rate (IMF), 2005 to 2015.

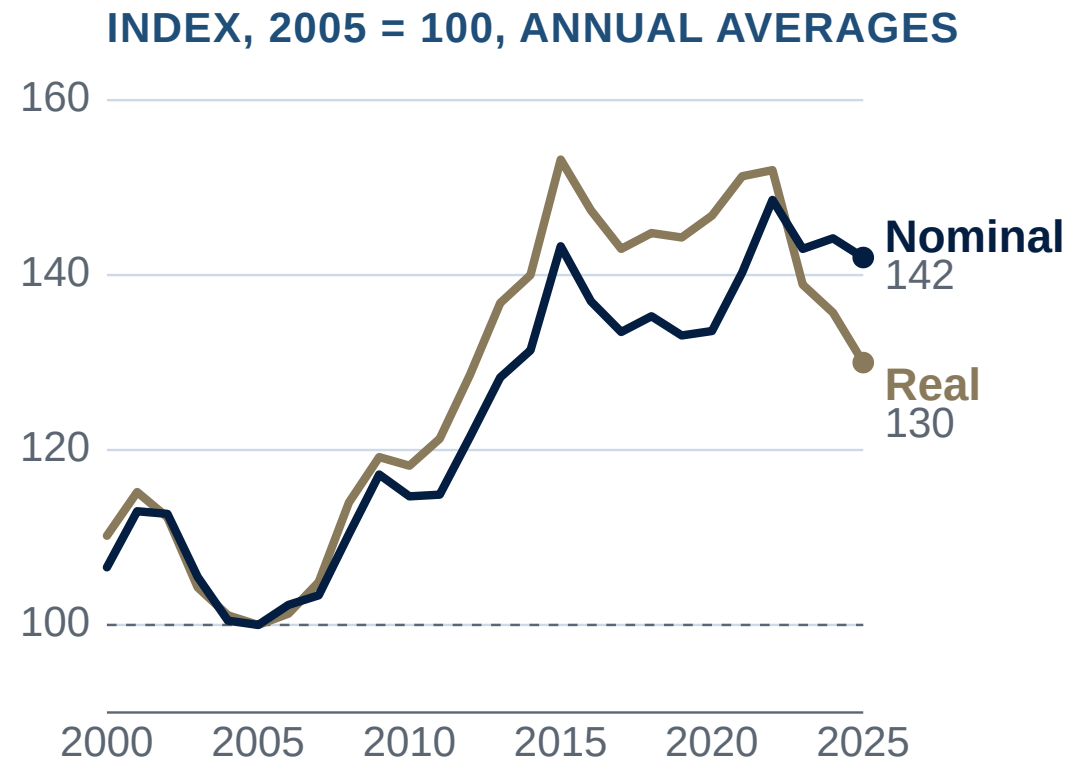
Fallows wrote that the RMB would rise, “and it will,” but that no rise would bring the jobs back to Ohio.

Location dependence is what it would cost a firm to do the activity anywhere else, given what has gathered around it.

The RMB rose by half in real terms, then fell back

Prices rose faster in China than abroad, so Chinese goods grew dearer faster than the currency alone shows.

Since 2022 slower inflation in China has made its goods cheaper again with the currency flat.



Bank for International Settlements, [Effective exchange rate indices](#), broad basket, nominal and CPI-based real, annual averages.

The real exchange rate is the nominal rate adjusted for prices at home and abroad, so it measures what Chinese goods cost in foreign goods.

The same cable sells at three prices and pays \$2

RETAIL PRICE OF THE SAME CABLE



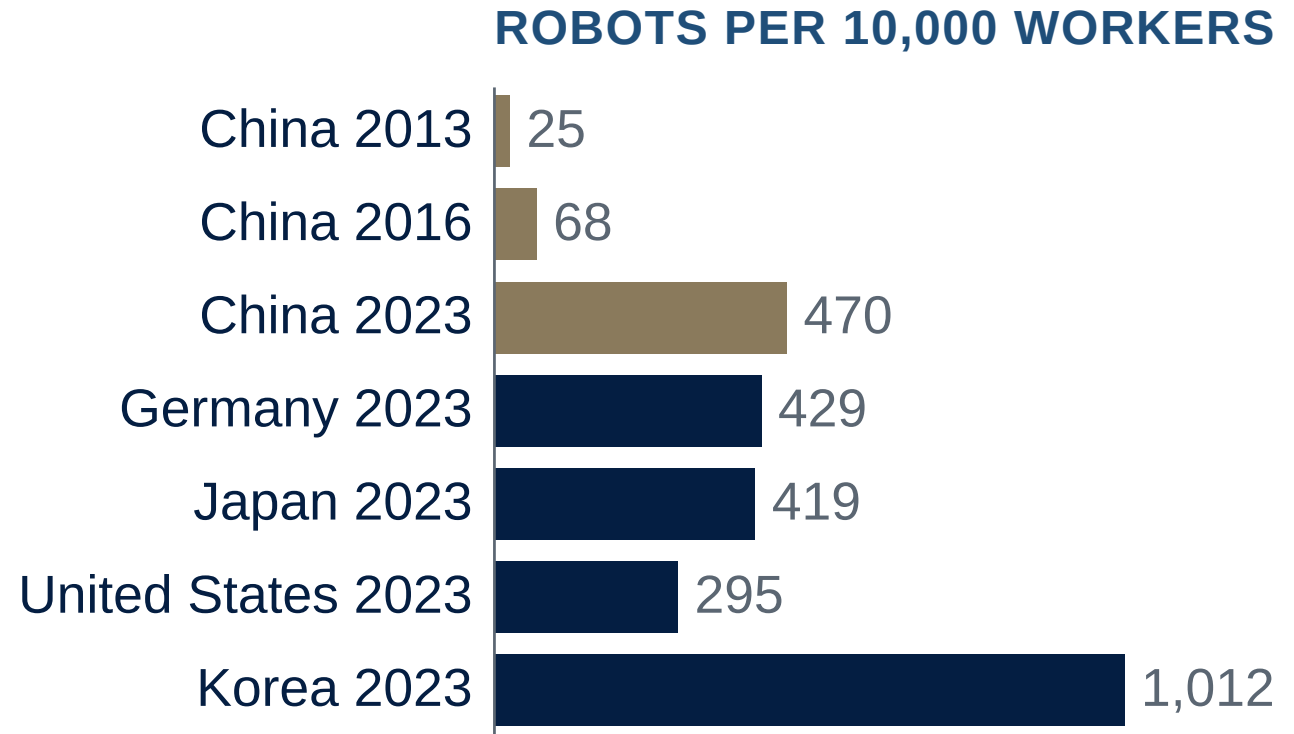
Prices as reported by Fallows (2007) for one cable sold under three labels.

Removing every subsidy would move the \$2 to \$3 or \$4, and neither the price nor the location would move with it.

As wages rose, China's factories bought robots

When labor stopped being cheap, the plants that stayed replaced workers with machines.

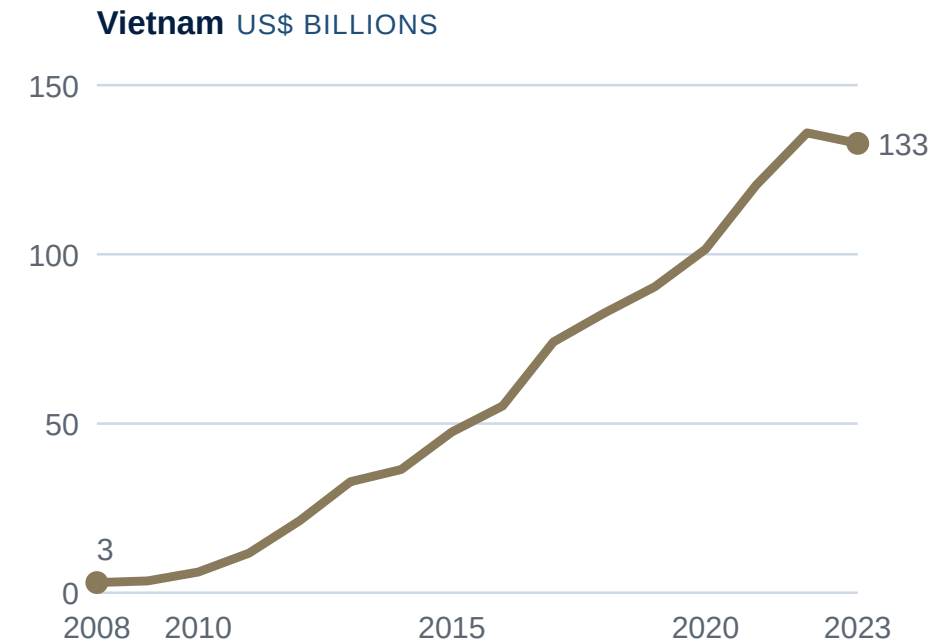
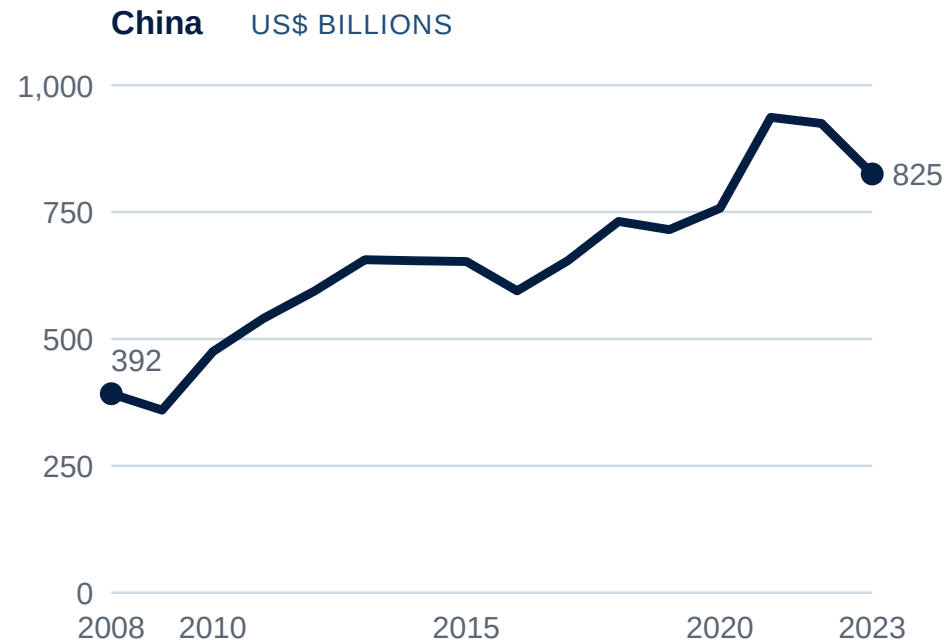
The assembly that needed the cheapest labor went elsewhere, and the rest stayed and automated.



International Federation of Robotics, [World Robotics](#), 2017 and 2024 editions.

As a place's inputs change, the activities it attracts change with them.

The next plant went to Vietnam



World Bank, [World Development Indicators](#), high-technology exports (UN Comtrade), 2008 to 2023.

Vietnam got the assembly stage, and most of the parts it assembles still come from China.

■ The labor-intensive stage moved on when China's labor stopped being cheap, and the supplier base stayed.

Group Activity: Locate One Activity in Your Industry

Locate one activity in your industry

1. **Select** one activity in your industry and the place where it concentrates.
2. **Diagnose** which input it uses most, why it is cheap there, and what has gathered around it.
3. **Recommend** where the next plant, office, or lab should go.
4. **Defend** the choice against a rival place that could copy the cheap input.

15 minutes • Industry teams • One annotated list

■ Report the input, the place, and what is slowest to rebuild elsewhere.

Why is your activity where it is, and what would be slowest to rebuild anywhere else?

5 minutes · Three reports, then one challenge from another team

The slowest thing to rebuild is what holds the activity in place.

Designed in California, assembled in China



Design sits where engineers are abundant and assembly where workers and suppliers are.

Raimond Spekking, [iPhone 5s \(model A1457\)](#), 2021 · Wikimedia Commons, CC BY-SA 4.0, cropped.

Where does an activity go, and what keeps it there?

An activity goes where the input it uses most is cheapest.

It moves only if its output travels for less than the cost gap.

Once firms gather, density becomes an input of its own.

When the cheap input moves, the next plant moves and the cluster stays.

On Wednesday we ask how one firm spans all these places.

What attracts production to the Pearl River Delta, and would your answer change if wages there doubled?

On Wednesday we read Landler and Barbaro on Wal-Mart and McGee's chapter 34.